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Fish herbivory as key ecological function in a heavily degraded coral reef system

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Abstract

Localized impacts, such as eutrophication and overfishing, have been identified as major threats to reefs worldwide, resulting in changes in coral reef community composition, from coral-dominance to algal-dominance. Herbivory by fishes is frequently cited as a key process structuring benthic community response to stressors. Here, we experimentally tested the capacity of herbivorous fishes to remove macroalgae along an onshore–offshore spatial gradient in the Spermonde Archipelago, Indonesia; a coral reef system exposed to long-term anthropogenic disturbances. Bioassays of *Sargassum* and *Padina* spp. were collected from the reef flat and transplanted onto the reef slope at five islands varying in distance (1–55 km) from the coastline and city of Makassar, South Sulawesi. Supplementary underwater video deployment and fish surveys were used to identify species-specific differences in herbivore communities and rates of algal removal. Over a 24 h period, *Padina* was almost completely removed from the two outermost islands, but removal declined toward shore. With the exception of the site closest to Makassar, more than 50% of all transplanted macroalgae were removed. Species diversity of algal-consuming fishes and total bites increased toward off-shore, with only one species, *Siganus virgatus* (Siganidae), feeding at all sites. These changes in herbivore communities underline the capacity for functional resilience under varying levels of reef degradation. Our results exemplify the complexities of ecological functioning along environmental gradients.

The process of herbivory is crucial for sustaining coral reef health (Hughes et al. 2003, 2007a; Mumby 2006; Burkepile and Hay 2006) by either directly or fortuitously removing benthic primary producers, particularly macro- and microalgae (Hughes et al. 2003; Littler et al. 2009). Fish and urchins which feed from the substratum can remove algae at early life stages or in established forms (Bellwood et al. 2004). It has been repeatedly shown through experimentation that, without herbivory, it is possible for algae to outgrow corals, thereby gradually changing reefs to a state of algal dominance (Hughes et al. 2007b; Smith et al. 2010). This is enhanced when increased nutrients and the removal of algae-consuming species occurs together, allowing algae to grow faster and to establish standing stocks that are able to outcompete corals for access to light and space. Given the continuous decline in the health of coral reefs around the world (Burke et al. 2011), ecological importance has been given to those fish species which are able to remove established algal communities, and return a reef to a state of coral dominance (Hughes et al. 2007b).

Fish communities on coral reefs are not constant in space or time (Warwick and Clarke 1993), and their structure can be a response to multiple biotic and abiotic factors (Warwick and Clarke 1993; Genner et al. 2010). However, fishing has been identified as particularly important, because it can remove traits, species and/or functions from a community, such as large predators (Jennings et al. 1995). Similarly, macroalgal communities can respond to changing water conditions (i.e., nutrients, sedimentation, temperature) and consumer communities (McCook 1999; Teichberg et al. 2010), altering their morphology and chemical composition. Unfortunately, for many reefs around the world, fishing and eutrophication, driven in particular by increasing coastal populations, are identified as the most common threats (Burke et al. 2011). Projections of further population growth suggest that these local stressors will become even more pressing in coming decades, with negative consequences for coral reef ecosystem integrity (Burke et al. 2011; Mora 2014). Herbivorous fishes on most reefs around the world are

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heavily impacted both directly and indirectly by overfishing (Edwards et al. 2014), suggesting that within these systems species richness and herbivore functioning may be increasingly diminished. Assessing how the composition of herbivorous fish communities and their ability to remove algae change in response to varying levels of reef degradation will help to understand the capacity of coral reefs to respond to projected impacts (Burkepile and Hay 2006).

Studies on Australia's Great Barrier Reef (GBR) have suggested that fish species capable of removing the brown macroalga *Sargassum* are of particular importance (Bellwood et al. 2006a) because this alga was dominant after long-term herbivore exclusion (Hughes et al. 2003). However, the dominant herbivorous fish species can change among seasons (Lefèvre and Bellwood 2011) and with differing spatial scales (Cvitanovic and Bellwood 2008; Hoey and Bellwood 2010; Bennett and Bellwood 2011; Vergés et al. 2012). Generally, species of the families Acanthuridae, Kyphosidae, Labridae (parrotfishes), and Siganidae are considered to be the most important herbivores, but the specific impact and role of the species can vary dramatically. Cross-continent comparison indicated that less than five species contributed to the greatest proportion of macroalgae removal on the GBR, while ~15 species where responsible at Ningaloo Reef (Vergés et al. 2012). These examples show that herbivory can be site- and time-specific and, furthermore, they also suggest a high level of ecological resilience within the herbivore functional group.

In the Indo-Pacific region, most studies examining macroalgal removal through the use of transplants have been conducted on the GBR. These studies have proven invaluable in understanding the potential processes that determine herbivory; however, their application for similar processes in other areas realizing greater levels of fishing may be limited. The cultural, political, and geographical setting of the GBR results in this reef being more impacted by large-scale processes (e.g. ocean acidification and temperature anomalies) than local fishing perturbations (Burke et al. 2011). Likewise, most reefs of the Indo-Pacific exhibit lower herbivore biomass than the GBR (Edwards et al. 2014) partly because of differences in fishing practices driven by local socioeconomic and governance conditions.

Indonesia's coral reefs, for example, are heavily impacted by changes in land and resource use. Agriculture, aquaculture, destructive coral reef resource use, ineffective law enforcement, and a large coastal population which heavily relies on coral reefs for food result in localized impacts being more pronounced than global processes (Burke et al. 2011, 2012). Given the location-specific threats which occur in the Southeast Asian region, and the high spatial and temporal variability of herbivory, it is of particular importance to examine the capacity for algal removal in coral reefs with high levels of localized impacts, such as that found in the Spermonde Archipelago, Southwest Sulawesi, Indonesia.

The Spermonde Archipelago is located within the Coral Triangle, an area which contains one of the most diverse assemblages of marine organisms in the world (Veron et al. 2009). This region also supports one of the largest reef fisheries in Indonesia. Furthermore, Spermonde features the second-largest port of the country, located at the city of Makassar (population: 1.4 million). The reefs of the area are exposed to heavy development of the coastal region, where land run-off and destructive fishing practices have existed for decades, resulting in poor water quality, and reef community and habitat degradation (Edinger et al. 1998; Pet-Soede and Erdmann 1998; Pet-Soede et al. 2001a,b). Related studies have found links with localized eutrophication and fishing impacts (Edinger et al. 1998, 2000; Pet-Soede et al. 2001a,b; Cleary et al. 2005; Becking et al. 2006; Cleary and Renema 2007; Knittweis et al. 2009; Sawall et al. 2011, 2013), the former being associated with increases in live coral with distance from Makassar (Edinger et al. 1998; Renema and Troelstra 2001). The direct effect of fishing practices on herbivores is unstudied, thus unclear, however, unselective fishing is intense (Pet-Soede et al. 2001a,b). This suggests possible impacts on herbivore communities and/or their preferred habitat, while near-shore water nutrient levels are high (Edinger et al. 1998; Sawall et al. 2011, 2012) for coral reefs. Despite a situation which would be expected to facilitate a change to algal dominance, macroalgal density is relatively low on the reef slope, and live coral cover is maintained even at the most impacted, near-shore sites (Sawall et al. 2013, Jompa unpubl.). No work has explored the role of herbivory in maintaining this dynamic system.

Therefore, this study aimed to experimentally and systematically assess in situ macroalgal removal rates and herbivore species composition across the Spermonde Archipelago. Transplanted macroalgal assays, underwater video, and underwater transect surveys were used to quantify the relative importance of herbivorous fish species in reducing macroalgae on the reefs and how these communities can change under severe ecological impacts, thereby aiming to provide insights on the importance of this key ecological function on impacted coral reefs.

Methods and Materials

Study Sites

This study was conducted between 8 and 25 September 2013, on five islands of the Spermonde Archipelago, Indonesia, along a transect of increasing distance from the city of Makassar (Fig. 1). Near-shore islands are affected by effluents from the city's harbor, sediments, aquaculture outflow, and wastewater from the fluvial discharge of nearby rivers (Renema and Troelstra 2001). Coral cover and diversity increases with distance from the mainland (Edinger et al. 1998), and the islands for this study were chosen to reflect

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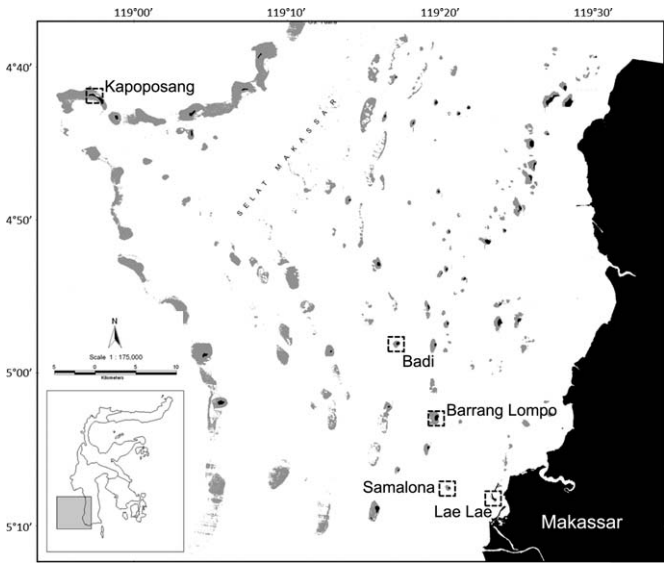

Fig. 1. Map of the Spermonde Archipelago. Dashed lines indicate study sites where bioassay experiments and surveys were conducted. Black indicates dry land and gray is reef.

these patterns along a cross-shelf gradient. Lae Lae (LL; 05°08 S, 119°23 E) was the closest (1 km distance) to the mainland, followed by Samalona (SA; 05°07 S, 119°20 E, 7-km distance), Barrang Lompo (BL; 05°02 S, 119°19 E, 11-km distance), Badi (BA; 04°57 S, 119°16 E, 19-km distance) and Kapoposang (KA; 04°41 S, 118°57 E, 55-km distance; Fig. 1).

All islands are located on the continental shelf except KA, which was the only site located on the edge of the continental shelf. Sampling was conducted during the dry season and there were little to no differences in weather conditions among days. The northwest corner of each island (except KA) was used to standardize the sampling sites among reefs. The western coast of the islands generally features a well-developed, carbonate fore-reef and a sandy back-reef and flat. The reef crest was shallow (~ 3 m) and the slope reaches down to 15 m. The last study site, KA, was located on the reef edge above the outer continental shelf wall (Fig. 1). Work at KA was conducted at the northeast site of the island at the edge of the carbonate shelf because this area displays the highest level of reef accretion.

Macroalgal transplants

Transplant bioassays and video cameras were used to assess species-specific rates of feeding on the macroalgae *Sargassum* sp. and *Padina pavonica*. *Padina* was selected as it represents one of the most abundant macroalgae on reef flats (Plass-Johnson unpubl.), while *Sargassum* is important in other comparatively similar areas (McClanahan et al. 2001; Hoey and Bellwood 2009), with the capacity to dominate coral reef communities after experimental phase shifts (Bellwood et al. 2006a; Hughes et al. 2007b).

All *Sargassum* and *Padina* transplants were collected from the same area of reef flat of BA throughout the experiment to control for among-reef differences in algae which may affect herbivore feeding (e.g., nutritional composition or secondary metabolites). Throughout the experiment, all samples were assumed to be the same species based on area of collection and similarity of morphology, however, species identification of *Sargassum sp*. was not feasible due to their poorly understood taxonomy (Kilar et al. 1992). Samples along with their holdfasts were chosen to represent their naturally occurring conditions (between 25 g and 118 g). Samples were collected daily and held in outdoor flow-through raceways at the field station on BL. Samples were spun for 10 s in a salad spinner and weighed to the nearest 1 g, labelled with a small cable tie, and then immediately returned to raceways. All samples were used for the experiment on the subsequent day after collection to minimize their deterioration.

At each reef for each day, six assays of each species were haphazardly deployed within a 200 × 2 m area on the reef slope at ~ 6-m depth. Five assays were attached with cable ties to artificial 60 × 60 cm steel anchors. The additional assay was placed in a 60 × 60 × 60 cm cage with 1-cm mesh to control for change in weight not due to herbivory (e.g., handling or in-water physical damage). Assays were deployed early in the morning (~ 07:00 h) and collected 24 h later. After collection, samples were again weighed after spinning to calculate weight loss due to herbivory. Transplants were conducted at each reef for 4 successive days at two concurrent reefs. BA and SA were completed first, followed by BL and LL, and lastly KA.

Video analysis

Video cameras (GoPro Hero3 in underwater housing) were mounted on steel rebar ~ 50 cm from one haphazardly chosen, uncaged, algal assay of each species at each reef on each day. The assays were filmed for approximately a 4-h period immediately after deployment (~07:30 until 11:30 h). The steel anchors that assays were attached to allowed for reference of fish size. Video recording was repeated daily, however, data from Day 1 was excluded (see Statistical Analyses section), totalling 12 h of video footage recorded per macroalga and island.

For each video, fish size (total length), number of bites, and species were recorded for each individual. Forays where it was not possible to discern between bites were recorded conservatively as a single bite. All fish observed feeding were recorded in size classes (2.5 cm for individuals < 10 cm, 5 cm for individuals > 10 cm, that is, bin: 0.1–2.5 cm, 2.6–5.0 cm, etc.) and then converted to kilogram based on published length–weight relationships (Kulbicki et al. 2005). To account for differences in fish sizes, the fish length was taken as the midpoint of each size class (Lefèvre and Bellwood 2011). To standardize for the impact of fish size, bite impact

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was calculated by multiplying fish size by bites (body mass in kg × bites; Cvitanovic and Bellwood 2008; Hoey and Bellwood 2010; Lefèvre and Bellwood 2011; Bennett and Bellwood 2011).

Herbivore survey

To evaluate differences in herbivore community composition, abundances and total biomass among islands, surveys (on SCUBA) were conducted 2 days prior to the start of and post each experiment. Roving herbivorous fishes of the families Acanthuridae, Kyphosidae, Labridae (parrotfishes), and Siganidae were counted and assigned to size classes (see Video Analysis section) using three 5 × 50 m transects. These fish families were selected because they have the greatest impact on benthic macroalgal communities (Ceccarelli et al. 2005). Similar to the transplants, fish survey transects were deployed 1–2 m below the reef crest. Swimming speed was constant and slow, with each transect taking ∼ 15 min. Counts were restricted to individuals > 3 cm to ensure identification was possible. Fish survey data were converted to biomass estimates based on published length–weight estimates.

Statistical analyses

For most of the islands, permutational analysis of variance (PERMANOVA) identified differences in tissue loss that were significantly less on Day 1 than successive days for both species of macroalgae (*Sargassum* PERMANOVA: Pseudo-$F_{15,80} = 8.58$, $p < 0.001$; *Padina* Pseudo-$F_{15,80} = 5.96$, $p < 0.001$), indicating that there was a period of time needed for herbivorous fishes to find and accept the assays during the experiment. Because of this, Day 1 was omitted from both the transplant and video analyses for all islands. There was no statistical difference among days after the removal of Day 1 (*Sargassum* PERMANOVA: Pseudo-$F_{10,60} = 0.45$, $p = 0.917$; *Padina* Pseufo-$F_{10,60} = 1.263$, $p < 0.284$), subsequently day was treated as the replicate ($n = 3$) for further analysis.

Weight loss for each alga was calculated similar to Bennett and Bellwood (2011). The mean proportional loss of mass (g) from the caged controls was used to estimate non-herbivore loss from assays. Therefore, mass loss was calculated as:

$$T_{\text{loss}} = 1 - H_a / (H_i \times [1 - C_a]),$$

where T_{loss} is the proportion of algae consumed, H_{a} is the algal mass (g) after 24 h exposure to herbivory, H_{i} is the initial algal mass, and C_{a} is the mean percentage lost from the controls (Table 2) of the associated island. Differences in consumed algae were analyzed among islands using a one-way PERMANOVA with site as a fixed factor. A Euclidean distance measure was calculated based on square-root transformed data (for normalization). Differences among islands were identified with pairwise comparisons using PERMANOVA. PERMANOVA was chosen because it is robust when

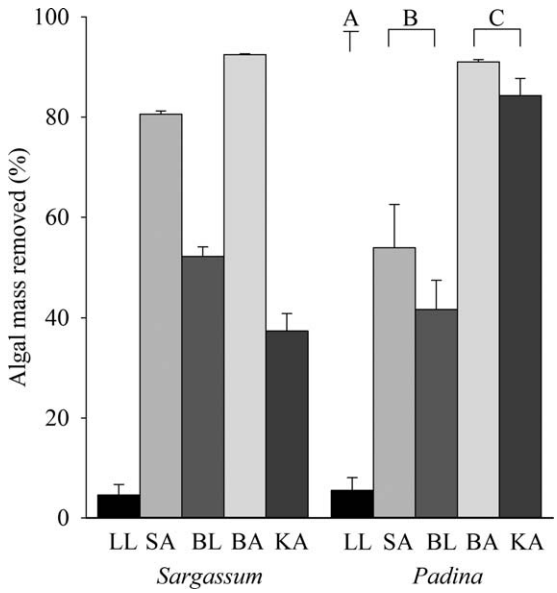

Fig. 2. Average (±SE) amount of *Sargassum* and *Padina* removed per 24-h period for consumer exposed treatment. Islands are in order from closest to furthest from Makassar (LL: Lae Lae, SA: Samalona, BL: Barrang Lompo, BA: Badi, KA: Kapoposang). Data represents total mass loss after the correction for controls. Data from Day 1 are not included. Lettering above *Padina* indicates islands that were not significantly different in the PERMANOVA pairwise post hoc test. There is not lettering above *Sargassum* because all islands were significantly different from each other.

the assumptions of normality and homogeneity of variance are not met (Anderson et al. 2008). This was the case given differences in macroalgae consumption among and within some reefs (e.g. BL), and high rates of removal (e.g. BA), leading to a highly negative skew. PERMDISP was used to test the assumption of homogeneity of variance (dispersion; Anderson et al. 2008), and in one case of significance (i.e., *Padina*, BA × LL, $p = 0.32$), interpretation was conservative. PERMANOVA with pairwise comparisons was used to identify difference in algal removal among islands.

The total contribution to mass-standardized bites, with a threshold of > 3% of total standardized bites at any given island, was used to identify species which had the greatest impact on macroalgae (*Sargassum* and *Padina* analyzed separately). Although a threshold of 3% is arbitrary, a similar threshold has been previously applied (Vergés et al. 2012) with the assumption that the impact of species which contributed to ≤ 3% was negligible. Data from these fish species were then used to characterize trends in herbivory across islands. Differences in herbivore biomass from underwater transect fish surveys were calculated with a one-way ANOVA with island as a fixed factor. Data were $\log(x + 1)$ transformed and checked for homogeneity of variance (Bartlett's test) and normality (using scatter plots).

All statistical analyses were performed using Primer-E v6 software (Clarke and Gorley 2006) with the PERMANOVA + add-on

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Table 1. PERMANOVA for test of Island after Day 1 was removed. All other days were pooled because they were not different from each other. Significant differences are indicated with bold.

	df	SS	MS	Pseudo-F	*P*(perm)	Uniq. Per.
Sargassum						
Island	4	1.22	0.31	250.09	**<0.001**	9917
Res	10	00.01	0.01			
Padina						
Island	4	1.19	0.30	39.30	**<0.001**	9929
Res	10	0.08	0.01			

package (Anderson et al. 2008). Fish species identification was based on Allen and Erdmann (2012).

Results

Algae removal

Removal of *Sargassum* was variable among islands, while the removal of *Padina* increased with distance from Makassar (Fig. 2). For *Sargassum*, removal was significantly different among islands (Table 1), with highest rates at BA and SA (Fig. 2). *Padina* removal was highest at the two outermost islands (BA and KA), intermediate at SA and BL, and lowest at LL (Fig. 2). Likewise, *Padina* removal was not significantly different between KA and BA, and between SA and BL (Table 1). Algal loss due to handling ranged from 2.3% to 6.7% (Table 2).

Video analysis

Across the islands, there was an increase in the total number of fish feeding on both macroalgae with increasing distance from Makassar, with the exception of a slight decrease from SA to BL for *Padina* (Table 3). This pattern was the same for the species which had the greatest impact in terms of standardized mass total bites (Table 3), with the total number of important herbivores being relatively low (≤ 5 species). Interestingly, through 24 h of video, there was not a single observation of feeding on either macroalga at LL. The average number of bites on *Sargassum* and *Padina* increased with distance from Makassar, except for a decrease in bites on *Padina*, again at BL (Table 3). The proportional contribution of the species that accounted for > 3% of total standardized bites was always above 94%, and above 99% for most islands (Table 3).

There were few fish species (contributing > 3% of mass-standardized bites) that overlapped among islands, and only one was found at all four (*Siganus virgatus*; Fig. 3A). *S. virgatus* fed on both macroalgae at all islands except for *Padina* at KK; it had more bites on *Sargassum* and was the most important consumer of *Sargassum* at SA and BL (Fig. 3). *S. margaritiferus* was the biggest consumer of both macroalgae at SA, but it did not occur in the videos at any of the other islands. Two acanthurids, *Naso unicornis* and *Naso lituratus*, were disproportionately important on the outer two reefs (Fig. 3). At BA, the former accounted for 83% and 95% of the standardized bites taken on *Sargassum* and *Padina*, respectively. At KA, *N. lituratus* accounted for 64% and 53% of the standardized bites on *Sargassum* and *Padina*, respectively.

Herbivore communities

From the visual surveys, herbivore biomass increased with increasing distance from Makassar, with the outer two islands (BA and KA) being different from the middle islands (SA and BL; ANOVA, $F_{4,1} = 320.4$, $p < 0.05$; Fig. 4). LL had significantly lower herbivore biomass than all other islands (Fig. 4). The parrotfishes were always proportionally the largest group of herbivores. The siganids (*Siganus*) were present at all islands, while the acanthurids (*Naso, Ctenochaetus, Zebrasoma*, and *Acanthurus*) became more abundant on the outer two islands (Fig. 4). In general, there were more fishes surveyed than those that fed on the assays, yet, there was little overlap in species (Table 4). Many of the important species in the video assay, for example *N. unicornis* and *N. lituratus*, were never observed in the visual surveys. *S. virgatus* was the only important species which occurred in both sets of data. Other universal species included *Scarus flavipectoralis* and *Siganus corallinus*, however, their contribution was not important in the videos.

Discussion

Our results indicate that in a heavily populated area with terrestrial influences, the coral reefs of the Spermonde Archipelago, the fundamental ecological function of herbivory is maintained by a few key, consumer fish species. These differences in algal removal across sites were reflected in distinct differences in the fish species involved in herbivory. The two furthest sites, BA and KA, exhibited more species feeding and more bites on the macroalgae than the sites closer to the mainland. The fish species at two of the three closest sites, SA and BL, had much less impact on the macroalgae during the time of video. At SA and BL, algae-consuming species were solely from the genus *Siganus*. Further off-shore

Table 2. Percentage (% ± SE) loss of algae from control assays. Any loss represents loss attributed to other disturbances than herbivory.

Island	*Sargassum*	*Padina*
Lae Lae	5.3%	6.7%
SE	(0.9%)	(0.7%)
Samalona	5.7%	8.0%
SE	(1.2%)	(1.2%)
Barrang Lompo	4.0%	6.0%
SE	(0.6%)	(1.5%)
Badi	2.7%	4.0%
SE	(1.3%)	(1.5%)
Kapoposang	2.3%	2.3%
SE	(1.3%)	(1.9%)

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Table 3. Total number of herbivore species feeding on transplanted *Sargassum* and *Padina* at each island with the number of species contributing >3% of total mass-standardized bites indicated within brackets. Total mass-standardized bites of all herbivore species and those which contributed to >3% of the total are given for each island. The proportion of total contribution of those species contributing to >3% of total mass-standardized bites is indicated in brackets. Islands are in order from closest to furthest from Makassar (LL: Lae Lae, SA: Samalona, BL: Barrang Lompo, BA: Badi, KA: Kapoposang).

			Total mass-standardized bites				Effect of algae
	Number of spp.		All species		Spp. contributing > 3%		
Island	*Sarg.*	*Padina*	*Sarg.*	*Padina*	*Sargassum*	*Padina*	
LL	0	0	0	0	0	0	
SA	5(2)	4(3)	265.9	37.2	263.9 (99%)	37.0 (99%)	0.671
BL	6(2)	3(2)	297.1	2.5	296.1 (99%)	2.4 (96%)	**0.034**
BA	6(3)	7(4)	599.6	721.4	598.5 (99%)	719.4 (99%)	0.793
KA	17(5)	11(4)	413.3	596.1	387.9 (94%)	592.8 (99%)	0.487

(BA and KA), the role of *Siganus* spp. was strongly overshadowed by the feeding of two *Naso* species (*N. unicornis* and *N. lituratus)* and one parrotfish, *Bolbometopon muricatum*.

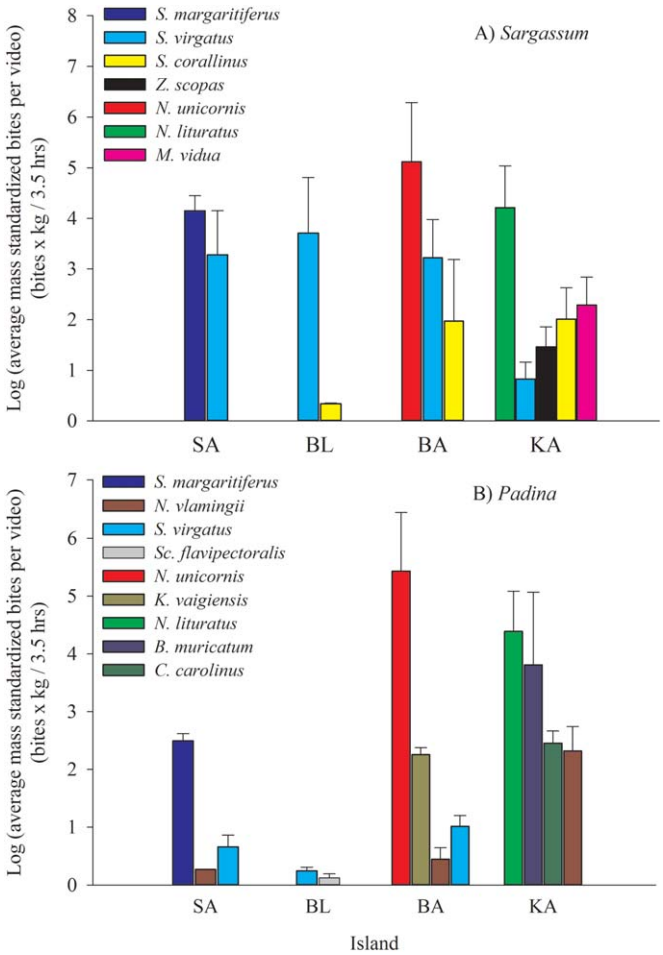

Fig. 3. Average number of mass standardized bites (±SE) from (A) *Sargassum* and (B) *Padina* assays by fish species across all islands. No fish were observed for Lae Lae on the videos.

Patterns in algal removal, particularly for *Padina*, showed marked spatial differences, increasing in relation to distance from shore. Patterns for *Sargassum* were more complex as feeding was more variable among sites, even with a decrease in its removal at the farthest site. Interestingly, at the site closest to Makassar (LL), there was no evidence of herbivory on either macroalga suggesting that herbivory in the Spermonde Archipelago is not maintained in an area with high human population densities.

Although the biomass (Wismer et al. 2009) and species diversity (Risk 1972) of macroalgal-consuming fishes has been correlated with live coral cover and rugosity (Wilson et al. 2006), discerning the specific cause of these relationships has been hard. Generally, near-shore reefs are exposed to higher levels of terrestrial effluents and fishing. Both can cause degradation of coral reef systems, but the relationship between the two, and their impacts on the benthos and fish communities can be ambiguous. Nonetheless, data from our

Table 4. Total number of herbivorous fish species counted during transect surveys. Also given are the species which occurred in both the video assays and transect survey.

Island	No. spp.	Species shared with video
Lae Lae	3	*S. flavipectoralis*
Samalona	15	*S. virgatus*
		S. flavipectoralis
Barrang Lompo	16	*S. virgatus*
		S. flavipectoralis
Badi	14	*S. corallinus*
		S. virgatus
		S. flavipectoralis
Kapoposang	17	*A. nigrofuscus*
		S. virgatus
		Z. scopas
		S. corallines
		N. vlamingii

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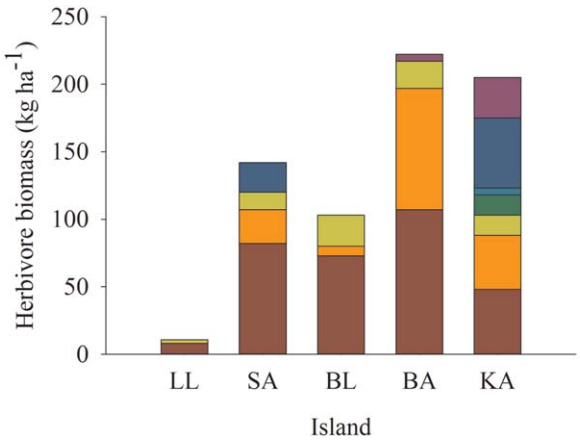

Fig. 4. Biomass of roving herbivores from five islands based on visual transect surveys. Groups were broken into their genera (red is *Scarus*, orange is *Chlorurus*, yellow is *Siganus*, green is *Zebrasoma*, teal is *Naso*, blue is *Ctenochaetus*, purple is *Acanthurus*). Data represent means, however, error bars have been omitted for better visualization. The largest standard error was for *Chlorurus* at BA ($\pm$ 37 kg ha^{-1}). Islands are ordered from closest (Lae Lae) to furthest from Makassar (Kapoposang).

transect surveys support the observations that herbivore diversity and biomass increase with an increase in distance from shore. Similar studies have linked on-shore to off-shore spatial gradients to herbivore biomass (Wismer et al. 2009) and theoretical functional metrics (Johansson et al. 2013), and our assay study supports these with evidence of actual herbivory processes. This pattern is also described by Hoey and Bellwood (2010) and Mantyka and Bellwood (2007), where herbivory was related to distance from shore on the GBR.

Browsing herbivorous fishes can vary in their ecological role both within and among species due to food preferences (Plass-Johnson et al. 2013) and behaviors (Fox and Bellwood 2011). Recent studies have found the siganids to be important herbivores (Fox and Bellwood 2008, 2011, 2012) with strong ecological differences within the group (Hoey et al. 2013). Hoey et al. (2013) identified dietary preferences in the group, with many species preferring brown foliose macroalgae over leathery brown algae. Similarly, the siganids are an important group of fishes in the Spermonde Archipelago, as they were the only group observed feeding at all reefs with the exception of LL. However, their feeding diversity and impact was much more intense on *Sargassum*, a brown leathery algae. *S. virgatus* was the only species seen feeding at each site. On the GBR, the sister species of *S. virgatus, S. doliatus* was also shown to be present at reefs across the continental shelf (Hoey et al. 2013). These observations suggest that only a few species may be robust in maintaining their function under changing environmental conditions, which lends importance to their conservation. However, the full ecological impact of the siganids is not completely clear as Fox and Bellwood (2008) note discrepancies between experimentally observed feeding and actual take-up of foods. Our observations cannot confirm the actual component ingested by the fishes, as we did not perform gut analysis as part of the study. However, siganids were directly observed removing and ingesting the algal assays. Previous studies on near-shore reefs of the GBR indicated that *N. unicornis* can be highly important in the removal of macroalgae (Hoey and Bellwood 2010; Lefèvre and Bellwood 2011), however we found this species, and the *Naso* group in general, to be nearly absent from the near-shore sites. These observations indicate regional plasticity in the diets of both the siganid and *Naso* species. We found a positive relationship for the removal of *Padina* and herbivore diversity, but not for *Sargassum*. In fact, the reef with the greatest diversity of consumers (KA) showed the second lowest biomass removal over 24 h for *Sargassum*, suggesting that diversity in herbivorous fishes may not directly relate to their capacity to remove algae and realized ecosystem services.

Increased input of nutrients and the removal of herbivores are continuously cited as the primary drivers of phase shifts on coral reefs (Burkepile and Hay 2006; Wilson et al. 2008; Hughes et al. 2010). In the Spermonde Archipelago, conditions have repeatedly been found to be eutrophic in the near-shore as far back as 1995 (Edinger et al. 1998), with suggestions of similar conditions well before these studies (Edinger et al. 1998, Sawall et al. 2011). The practice of destructive fishing techniques in the Spermonde Archipelago is well noted in the literature (e.g. Pet-Soede and Erdmann 1998; Sawall et al. 2013), however, reporting of their effects is generally limited to changes in the benthic communities. In this study, however, we did not test direct links between nutrients and/or fishing practices with primary production or rates of herbivory. Rather, historical records of nutrient input and destructive fishing provide insight into the processes resulting in the current spatial patterns seen in the coral reef community composition of the archipelago. As such, the cross-shelf increase in herbivore diversity and biomass in the Spermonde Archipelago may be a product of anthropogenic processes, but the direct link is unclear. The increased observations of herbivorous species, for example of *Naso* and *Bolbometopon*, at off-shore sites may reflect either environmental impacts or fishing practices, as these fishes tend to be vulnerable to heavy fishing practices (Bellwood et al. 2012; Bejarano et al. 2013), spatial changes in preferred habitat, or a combination of both factors.

One of the primary concepts of ecological resilience is the capacity of systems to reorganize while undergoing change, yet retaining the same function (Walker et al 2004). While the community composition of algal-feeding fishes, with the exception of one species with a low proportion of overall bites (*S. virgatus*), differed across sites, the herbivore communities were able to fulfil the function of macroalgal maintenance. The presence of *B. muricatum*, thought to be a bioeroder (Hoey and Bellwood 2009), at KA demonstrates that previously unknown species can also contribute to the important herbivory process (Bellwood et al. 2006b; Plass-Johnson et al. 2014), while variation in feeding by the siganids and *Naso* species exemplify the potential for resilience

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within the herbivore function. The situation was much different at LL, the site closest to Makassar. The transplant experiment suggested that herbivory does not play a significant role in controlling macroalgae at this site. Alternatively, anecdotal evidence from the videos showed pieces of *Sargassum* breaking off from the bioassays. These observations suggest that, given the island's near-shore position, reduced light attenuation and sedimentation may play a larger role than herbivory in maintaining low macroalgae abundances via physiological and physical stress (Kleypas et al. 1999).

It should not be overlooked that this was an experimental situation where fully grown macroalgae were offered to consumers. Our observed differences between underwater transect and video methods probably directly reflect the experimental offering of foods. This likely attracted species to the area, which was previously devoid of these algae. Discrepancies between video and diver-based surveys are well noted in the literature (e.g. Mallet et al. 2014), nevertheless, observed patterns of the presence (transect survey and video) and impact (video) of herbivores both positively correlate with live coral and rugosity, suggesting that a healthier coral reef habitat supports a greater variety of herbivorous species. Similarly the experimental transplanting of macroalgae does not take into account all processes controlling macroalgal removal during the entire algal life cycle, which may involve different herbivore species or density-dependent processes that may affect herbivory. A further caveat of our experiment is that we were not able to assess possible urchin feeding at night. However, evidence from the videos, and in some cases, the complete removal of full algal assays within the recorded 4.5 h, strongly suggest that fish herbivory accounted for the majority of the algal loss.

Nonetheless, our study shows that, in a system exposed to long-term anthropogenic disturbance, herbivores were able to remove the majority of both macroalgae over a 24-h period at all sites, failing to do so only when impacts became extremely strong. Capacity for herbivory and its resilience is perhaps facilitated by the high regional species diversity allowing for the functional replacement of species under varying levels of impacts. Even though the identity of herbivores contributing to algal removal in this study was highly variable along the cross-shelf gradient, herbivory was sustained throughout the system except for the site of the highest localized impacts, demonstrating remarkable resilience in herbivory on the scale of the shelf system.

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